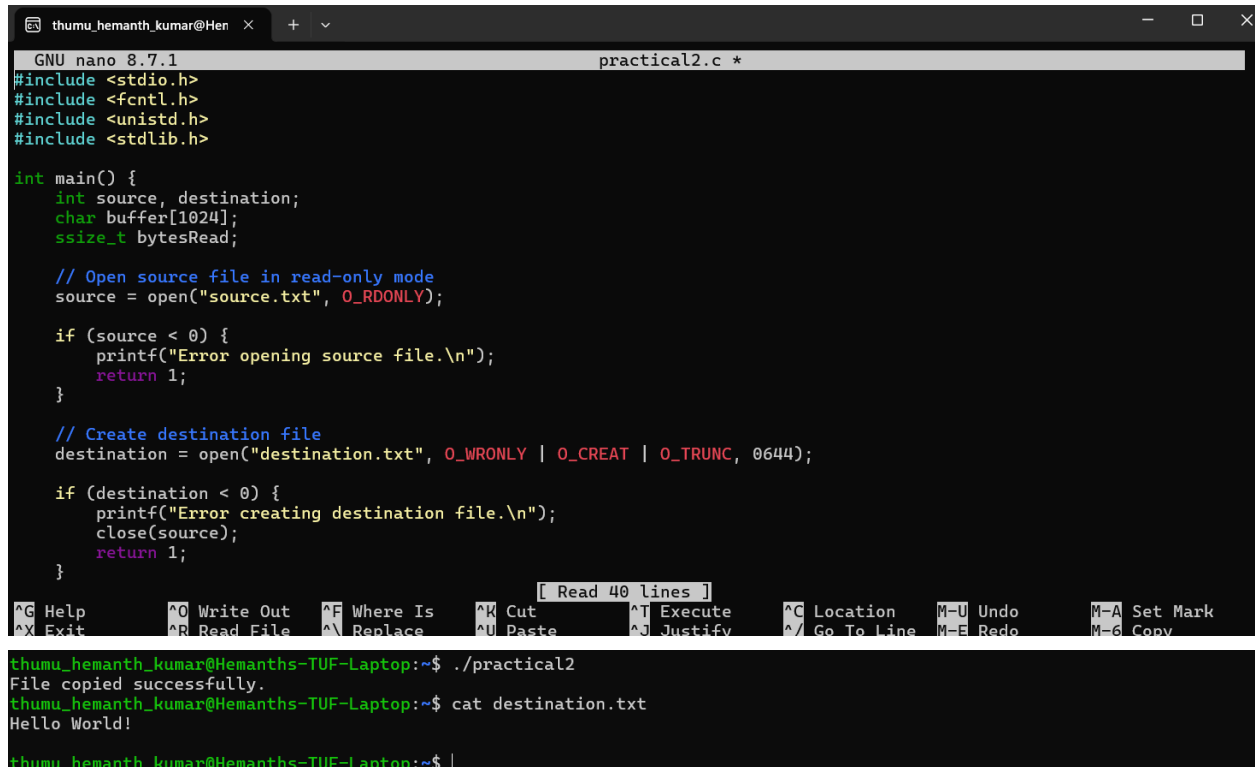


2. Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

Use the `strace` utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved.



The screenshot shows a terminal window with a nano editor editing a file named `practical2.c`. The C code defines a `main` function that opens `source.txt` in read-only mode, checks for errors, creates `destination.txt` in write-only mode with `O_CREAT` and `O_TRUNC` flags, checks for errors, and then copies the contents of `source.txt` to `destination.txt` using a loop with `read` and `write` system calls. After the copy, it closes both files and returns 0. Below the editor, the terminal shows the execution of `./practical2`, which outputs "File copied successfully.", followed by `cat destination.txt`, which outputs "Hello World!".

```
GNU nano 8.7.1 practical2.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main() {
    int source, destination;
    char buffer[1024];
    ssize_t bytesRead;

    // Open source file in read-only mode
    source = open("source.txt", O_RDONLY);

    if (source < 0) {
        printf("Error opening source file.\n");
        return 1;
    }

    // Create destination file
    destination = open("destination.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (destination < 0) {
        printf("Error creating destination file.\n");
        close(source);
        return 1;
    }

    while ((bytesRead = read(source, buffer, 1024)) > 0) {
        write(destination, buffer, bytesRead);
    }

    close(source);
    close(destination);
    return 0;
}

thumu_hemanth_kumar@Hemanths-TUF-Laptop:~$ ./practical2
File copied successfully.
thumu_hemanth_kumar@Hemanths-TUF-Laptop:~$ cat destination.txt
Hello World!
thumu_hemanth_kumar@Hemanths-TUF-Laptop:~$
```